

Figure 3

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1 Packages, data and functions

1.1 Packages

```
library(tidyverse)
library(here)
library(survey)
library(lavaan)
library(lavaan.survey)

# lavaan.survey is an extension of lavaan that can deal with data independency.
# It is no longer on cran, but it is a trustworthy package (there is also a
# publication for it). The source code is
# here https://github.com/daob/lavaan.survey. To get the package you need to
# clone the repository linked above and then build the package locally.
```

1.2 Data

```
complete.forecasts <- readRDS(here("Data/complete.forecasts.RDS"))

semDD.simplex <- complete.forecasts %>%
  dplyr::filter(forecast.method == "Simplex EDM",
                Target != "Total_biomass",
                Target != "Mean_oxygen") %>%
```

```
ungroup()
```

2 Figure 3

2.1 Models

```
model <-  
"  
RMSE ~ median_richness + cv.res + pe5 + acf_lag5  
cv.res ~ median_richness  
pe5 ~ median_richness  
acf_lag5 ~ median_richness  
cv.res ~~ pe5  
"  
  
## create random effect variable  
semDD.simplex$group <- paste0(semDD.simplex$Target, "_",  
                              semDD.simplex$bottle, "_",  
                              semDD.simplex$incubator)  
  
# semDD.simplex, semDD.rf, semDD.mv, semDD.rnn, semDD.ar  
out <- sem(model, data=semDD.simplex, group="light_treatment")  
  
### Survey Design Object  
des.card <- svydesign(ids = ~ group, nest=T, data = semDD.simplex)  
  
### Post-Processing  
fit.out.survey <- lavaan.survey(out, des.card, estimator = "MLM")  
  
summary(fit.out.survey, standardized=T, rsquare=T)  
  
## lavaan 0.6.17 ended normally after 96 iterations  
##  
##      Estimator                      ML  
##      Optimization method          NLMINB  
##      Number of model parameters    32
```

```

##
## Number of observations per group:
## Group 1 126
## Group 2 127
##
## Model Test User Model:
## Standard Scaled
## Test Statistic 3.219 3.270
## Degrees of freedom 4 4
## P-value (Chi-square) 0.522 0.514
## Scaling correction factor 0.984
## Satorra-Bentler correction
## Test statistic for each group:
## Group 1 2.977 3.024
## Group 2 0.242 0.246
##
## Parameter Estimates:
##
## Standard errors Robust.sem
## Information Expected
## Information saturated (h1) model Structured
##
##
## Group 1 [Group 1]:
##
## Regressions:
## Estimate Std.Err z-value P(>|z|) Std.lv Std.all
## RMSE ~
## median_richnss 0.015 0.031 0.469 0.639 0.015 0.035
## cv.res -0.602 0.177 -3.412 0.001 -0.602 -0.416
## pe5 0.085 0.659 0.128 0.898 0.085 0.017
## acf_lag5 -0.695 0.112 -6.230 0.000 -0.695 -0.368
## cv.res ~
## median_richnss 0.005 0.022 0.243 0.808 0.005 0.018
## pe5 ~
## median_richnss 0.010 0.007 1.459 0.145 0.010 0.117

```

```

##   acf_lag5 ~
##   median_richnss   -0.018    0.019   -0.948    0.343   -0.018   -0.081
##
## Covariances:
##           Estimate Std.Err z-value P(>|z|) Std.lv Std.all
##   .cv.res ~~
##   .pe5           -0.032    0.008   -4.110    0.000   -0.032   -0.784
##
## Intercepts:
##           Estimate Std.Err z-value P(>|z|) Std.lv Std.all
##   .RMSE           1.177    0.474    2.485    0.013    1.177    2.151
##   .cv.res           0.416    0.119    3.486    0.000    0.416    1.101
##   .pe5              0.595    0.036   16.347    0.000    0.595    5.471
##   .acf_lag5         0.517    0.103    5.012    0.000    0.517    1.782
##
## Variances:
##           Estimate Std.Err z-value P(>|z|) Std.lv Std.all
##   .RMSE           0.203    0.041    5.006    0.000    0.203    0.678
##   .cv.res           0.143    0.034    4.151    0.000    0.143    1.000
##   .pe5              0.012    0.002    5.389    0.000    0.012    0.986
##   .acf_lag5         0.084    0.006   13.688    0.000    0.084    0.993
##
## R-Square:
##           Estimate
##   RMSE           0.322
##   cv.res           0.000
##   pe5              0.014
##   acf_lag5         0.007
##
##
## Group 2 [Group 2]:
##
## Regressions:
##           Estimate Std.Err z-value P(>|z|) Std.lv Std.all
##   RMSE ~
##   median_richnss   -0.077    0.035   -2.224    0.026   -0.077   -0.220

```

```

##      cv.res          -0.095    0.173   -0.548    0.583   -0.095   -0.072
##      pe5             0.430    0.569    0.756    0.449    0.430    0.092
##      acf_lag5        -0.350    0.143   -2.455    0.014   -0.350   -0.185
##      cv.res ~
##      median_richnss  -0.030    0.026   -1.150    0.250   -0.030   -0.111
##      pe5 ~
##      median_richnss   0.014    0.007    2.186    0.029    0.014    0.191
##      acf_lag5 ~
##      median_richnss   0.010    0.017    0.592    0.554    0.010    0.053
##
## Covariances:
##              Estimate Std.Err z-value P(>|z|) Std.lv Std.all
##      .cv.res ~~
##      .pe5          -0.022   0.006  -3.968   0.000  -0.022  -0.628
##
## Intercepts:
##              Estimate Std.Err z-value P(>|z|) Std.lv Std.all
##      .RMSE          1.055   0.394   2.681   0.007   1.055   2.235
##      .cv.res          0.713   0.143   4.978   0.000   0.713   1.982
##      .pe5             0.540   0.037  14.714   0.000   0.540   5.319
##      .acf_lag5        0.506   0.093   5.454   0.000   0.506   2.026
##
## Variances:
##              Estimate Std.Err z-value P(>|z|) Std.lv Std.all
##      .RMSE           0.201   0.041   4.898   0.000   0.201   0.903
##      .cv.res          0.128   0.037   3.444   0.001   0.128   0.988
##      .pe5             0.010   0.001   8.040   0.000   0.010   0.963
##      .acf_lag5        0.062   0.007   9.524   0.000   0.062   0.997
##
## R-Square:
##              Estimate
##      RMSE          0.097
##      cv.res         0.012
##      pe5            0.037
##      acf_lag5       0.003

```

2.2 Figure

Figure 3 was done with MS PowerPoint and the above model.

3 Versions of R and of the R-packages used

R version: R version 4.3.2 (2023-10-31).

Base packages: grid, stats, graphics, grDevices, utils, datasets, methods, base.

Other packages: lavaan.survey 1.1.3, lavaan 0.6-17, survey 4.2-1, survival 3.5-7, Matrix 1.6-5, here 1.0.1, lubridate 1.9.3, forcats 1.0.0, stringr 1.5.1, dplyr 1.1.4, purrr 1.0.2, readr 2.1.5, tidyr 1.3.1, tibble 3.2.1, ggplot2 3.5.0, tidyverse 2.0.0, knitr 1.45.

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